Core Focus on Proportions & Probability

Block 3 Review ~ Percents

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. Wren wrote some pairs of numbers she thinks are equal. Decide which pairs really are equal. Circle all that apply.

A. 1.8 = 18%

B. 0.006 = 0.6%

C. $\frac{7}{8}$ = 87.5%

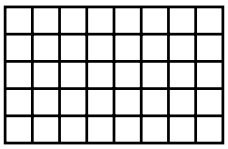
D. 36% = $\frac{9}{25}$

E. $2\frac{1}{4}$ = 22.5%

F. 0.3 = 3%

B, C, D. (D: 36/100 = 9/25. Wrong: A: 1.8 = 180%. E: 2 1/4 = 2.25 = 225%. F: 0.3 = 30%.)

2. A grid has 5 rows of 8 equal squares, for 40 squares in all. How many squares must be shaded to show 35% of the grid?



A. 14 B. 20 C. 26 D. 35

A. 35/100 = *x*/40 → 100*x* = 1,400 → *x* = 14 squares

3. Ellie set a goal to read 24 books this year. She read 125% of her goal. How many books did Ellie read?

A. 6 B. 19.2 C. 25 D. 30

D. 1.25 × 24 = 30 books (more than 100% of the goal, so more than 24)

4. What is the percent decrease from 45 to 29? Round to the nearest tenth of a percent.

A. 16% B. 35.6% C. 55.2% D. 64.4%

B. Change = 16; 16 ÷ 45 = 0.3555… ≈ 35.6% (divide by the original amount, 45, not 29)

5. A theater sold 320 tickets on Friday. On Saturday it sold 20% fewer tickets. How many tickets did it sell on Saturday?

A. 64 B. 240 C. 256 D. 384

C. 0.2 × 320 = 64 fewer; 320 − 64 = 256 tickets

6. Tasha earns a 3% commission on everything she sells. Last week she sold $1,240 worth of furniture. How much commission did she earn?

A. $3.72

B. $37.20

C. $372.00

D. $1,277.20

B. 0.03 × 1,240 = $37.20

7. Rosa paid $18.60 for a pair of boots that were on sale for 40% off. What was the original price of the boots?

A. $7.44

B. $11.16

C. $26.04

D. $31.00

D. 40% off means she paid 60%: 0.6*x* = 18.60 → *x* = 18.60 ÷ 0.6 = $31.00

8. The Okafor family's dinner bill was $56.00. They left an 18% tip. For numbers 8a – 8d, determine whether each statement about their payment is true or false.

8a. The tip was $5.60.	TRUE	FALSE
8b. The tip was $10.08.	TRUE	FALSE
8c. The family paid more than $56.00 in all.	TRUE	FALSE
8d. The family paid $65.08 in all, including the tip.	TRUE	FALSE

8a FALSE ($5.60 is only 10%) · 8b TRUE (0.18 × 56 = $10.08) · 8c TRUE (a tip is added) · 8d FALSE ($56.00 + $10.08 = $66.08)

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Part II – Constructed Response

Write each number in the form asked.

1. Write 64% as a fraction in simplest form and as a decimal.

64/100 = 16/25 (divide by 4); 0.64

2. Write $1\frac{3}{8}$ as a percent.

137.5% (3 ÷ 8 = 0.375, so 1 3/8 = 1.375; × 100)

Solve. Show all work necessary to justify your answer.

3. What is 35% of 260?

91 ($x = 0.35 \times 260$)

5. What percent of 120 is 3?

2.5% ($3 = x \cdot 120 \rightarrow x = 0.025$)

4. Twenty-one is 60% of what number?

35 ($21 = 0.6x \rightarrow x = 21 \div 0.6$)

6. Felix was asked, "Forty-five is 75% of what number?" His work is below. Explain his mistake and find the correct answer.

$x = 0.75 \cdot 45$
$x = 33.75$

45 is the part, not the whole. Correct: $45 = 0.75x \rightarrow x = 45 \div 0.75 =$ **60**. (Check: $0.75 \times 60 = 45$.)

Find each selling price. Use mathematics to justify your answer.

7. original price: \$64.00
percent of markup: 35%

\$86.40 (markup $0.35 \times 64 =$ \$22.40; \$64.00 + \$22.40)

8. original price: \$52.00
percent of discount: 15%

\$44.20 (discount $0.15 \times 52 =$ \$7.80; \$52.00 − \$7.80)

9. Imani asked 40 students how they get to school. The results are in the table. Find the percent of students for each response. Show all work necessary to justify your answer.

Way to school	Bus	Car	Walk	Bike
Number of students	18	12	7	3

Bus 18/40 = 45% · Car 12/40 = 30% · Walk 7/40 = 17.5% · Bike 3/40 = 7.5% (total 100%)

10. A video game costs \$35.00. It is on sale for 20% off. Then an 8% sales tax is added to the sale price. How much does the game cost in all? Show all work necessary to justify your answer.

\$30.24. Sale price: \$35.00 − $0.2 \times 35 =$ \$28.00. Tax: $0.08 \times 28 =$ \$2.24. Total: \$28.00 + \$2.24 = \$30.24.

11. Kai bought a skateboard. Its original price was $85.00, and it was on sale for 30% off. Kai did the work shown below to find the sale price. Show another way to find the price Kai paid.

Let x = the discount

$x = 85 \cdot 0.30$
$x = 25.50$
The discount is $25.50.

Sale price:
$85.00 − $25.50 = $59.50

I will pay $59.50 for the skateboard.

$59.50. Sample: 30% off means he pays 100% − 30% = 70% of the price, so 0.7 × 85 = $59.50. Or a proportion: 70/100 = x/85 → x = 59.50.

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Part III – Problem Solving

A store pays $80 for a lamp. It marks the price up 60% to set its selling price. Later, the lamp goes on sale for 25% off the selling price. Maya says the sale price is 35% more than the $80 the store paid, because 60% − 25% = 35%. Is Maya correct? What is the overall percent of change from the $80 the store paid to the sale price? Use mathematics to justify your answer.

Selling price: 0.6 × 80 = $48 mark-up, so $80 + $48 = $128. Sale: 0.25 × 128 = $32 off, so $128 − $32 = $96. Change: $96 − $80 = $16 increase; 16 ÷ 80 = 0.20 = **20% increase**. Maya is **not** correct (35% would be $108). The 25% is taken off $128, not off $80. Check: 1.6 × 0.75 = 1.2, so the sale price is 120% of $80.